

ORIGINAL ARTICLE

Open Access



Low-cost negative emissions by demand-side management for adsorption-based direct air carbon capture and storage

P. Postweiler^{1*} , D. Rezo¹ , M. Engelpracht¹ , B. Nilges¹ and N. von der Assen^{1*}

Abstract

Limiting anthropogenic climate change to below 2 °C requires substantial and rapid reductions in greenhouse gas emissions. Additionally, carbon dioxide removal technologies are essential to compensate for hard-to-abate emissions and counteract overshooting the earth's carbon budget. One prospective technology is direct air carbon capture and storage (DACCS), but its energy intensity and costs limit large-scale deployment. Flexible DACCS operation seems promising for cost reduction but yet remains underexplored. This study explores the economic benefits of flexible operation of adsorption-based DACCS, considering fluctuations in both electricity prices and greenhouse gas emissions from the electricity supply. To increase the feasibility of flexible DACCS operation, the typical steam-assisted temperature vacuum swing adsorption cycle is enhanced by introducing two break phases and variable air and steam mass flows during adsorption and desorption. The benefits of flexible operation are comprehensively evaluated using a DACCS system model integrating a detailed dynamic process model with life-cycle greenhouse gas emissions and economic data. The flexible operation allows each cycle to be adjusted to optimally address the time-varying greenhouse gas emissions and costs from electricity supply. A rolling horizon algorithm combined with particle swarm optimization is used to optimize the DACCS cycles in flexible operation mode over one week. The case study focuses on the future German power grid and a DACCS system using amine-functionalized sorbents. Results indicate that flexible DACCS operation can significantly reduce net carbon removal costs by up to 20 % compared to a steady-state operation. These findings highlight the potential of flexible DACCS operation to support carbon neutrality efforts by enabling cost-effective carbon dioxide removal through integration with volatile renewable energy systems.

Keywords Negative emission technologies, Flexible operation, Life-cycle GHG emissions, Net removal costs, Dynamic simulation and optimization

1 Introduction

Limiting anthropogenic climate change to below 2 °C is a significant challenge for humanity. Foremost, greenhouse gas (GHG) emissions must be substantially and rapidly reduced [1]. In addition, carbon dioxide removal (CDR) methods are mandatory according to almost all modeled emission pathways of the IPCC that reach the 2 °C goal [1]. CDR is required to compensate for GHG emissions that are hard to abate or unavoidable, as well as to counteract overshooting the earth's carbon budget [1–3].

*Correspondence:

P. Postweiler
patrik.postweiler@itt.rwth-aachen.de
N. von der Assen
niklas.vonderassen@itt.rwth-aachen.de

¹ Institute of Technical Thermodynamics, RWTH Aachen University, Aachen 52062, Germany

CDR methods range from nature-based approaches like afforestation or peatland restoration to technological solutions such as bioenergy with carbon capture and storage or direct air carbon capture and storage (DACCS) [4]. Each method has its advantages and challenges [5, 6], and no single method can achieve the required level of negative emissions alone [7]. Hence, a diverse set of CDR methods is imperative.

Due to its high scalability, reasonable technology readiness level, and long-term storage capabilities, DACCS is a promising CDR method [4]. DACCS involves capturing CO₂ directly from the atmosphere – referred to as direct air capture (DAC) – and storing it permanently. Various DAC processes exist that are based on different physical mechanisms, e.g., adsorption, absorption, electrochemistry, or membranes [7–12]. All DAC options are energy-intensive, making DACCS an energy- and cost-intensive CDR method. The energy intensity necessitates the use of low-carbon energy sources. Otherwise, substantial CO₂ emissions from the energy supply could lead to high net removal costs or even result in more CO₂ being emitted than captured [13–15]. Consequently, low-carbon energy sources are imperative for DACCS.

Low-carbon electric power systems typically have a high share of volatile renewables like wind or solar. A high share of these volatile renewables increases the chance of temporal mismatches between electricity supply and demand [16]. In fossil-based electric power systems, the supply side can be adjusted by managing power plants [17] to counter a mismatch between supply and demand. However, managing the supply side becomes insufficient with an increasing share of volatile renewable electricity sources.

Alternatively, demand-side management (DSM) can increase the flexibility of the demand by shifting, throttling, or ramping up the load [18, 19]. From a broad energy-system perspective, DSM could increase grid stability, reduce overall costs, and decrease GHG emissions [18, 20]. The importance of DSM increases as larger shares of volatile renewables are integrated into the energy system. The higher the share of renewables, the more valuable DSM becomes for maintaining grid stability and cost efficiency. From an energy-system perspective, the specific process providing flexible demand is less important than ensuring sufficient flexible demand within the system. To ensure sufficient flexible loads within an energy system, strong financial incentives, such as temporarily low electricity prices or market-based compensation mechanisms, are crucial for encouraging process operators to provide the needed demand flexibility.

Applying DSM to DACCS appears to be a promising operational strategy for two reasons. First, DACCS

could particularly benefit from temporarily low electricity prices due to its high energy demand. Second, the climate benefit of DACCS could be increased when using a flexible operation that exploits variability in the CO₂-intensity of the power supply [14]. Consequently, DACCS is seen as a promising flexible load for volatile electric power systems [21–24].

Singh and Colosi demonstrated the potential of flexibly operated DACCS that utilizes excess renewable electricity [22]. Their findings indicate that DACCS can remain economically viable even with capacity factors as low as 30 % [22]. Further, Alabi et al. examined the optimal real-time scheduling of DACCS alongside other CDR technologies within an energy system [25]. They revealed that flexible DACCS operations are favorable within optimally scheduled frameworks [25].

Even though the advantages of flexible DACCS operation are well-documented [21, 22, 24, 25], the specific mechanisms in the DACCS process by which this flexibility is achieved remain largely unaddressed [26]. For absorption-based DACCS, Arwa and Schell implemented flexibility using batteries and silos [26]. Although silos proved to be more cost-effective than battery storage in the absorption-based DACCS system [26], silos are not universally applicable to all DACCS technologies. For adsorption-based DACCS, Wiegner et al. outlined flexibility options by integrating batteries and adaptations to the DACCS process itself [27]. Furthermore, Liu et al. outlined flexibility options through the use of dispatchable DACCS modules and flexible adaptations of the adsorption phase time [28]. Liu et al. showed the potential to utilize unused curtailed wind power and to even earn a profit with flexible DACCS in future scenarios [28].

These pioneering studies started to explore how DACCS systems can operate more flexibly, but significant research gaps still remain. Particularly, for flexible operation of adsorption-based DACCS, adequate control algorithms are lacking [27]. Considering that adsorption-based DACCS is the most technologically advanced option [8], it is essential to address these gaps. Therefore, this work aims to expand knowledge in the flexible operation of adsorption-based DACCS to enhance DSM capabilities. In particular, this work addresses three research gaps that are still pending to the best of the authors' knowledge:

- (i) Limited suggestions have been made on how the standard cycles for adsorption-based DACCS have to be adjusted to enhance flexible operation without using expensive energy storage units. Standard cycles are temperature vacuum swing adsorption (TVSA) and steam-assisted temperature vacuum swing adsorption (S-TVSA).

- (ii) No analysis has been performed on the flexible operation of an adsorption-based DACCS system based on detailed dynamic process models. Such detailed dynamic models capture the transient and discontinuous behavior, which greatly impacts the overall performance.
- (iii) No adsorption-based DACCS system model has been proposed that allows for a holistic evaluation of the net carbon removal cost when considering flexible operation.

This study proposes adaptations to the standard cycle of adsorption-based DACCS to enhance flexibility without using energy storage, as presented in Section 2. Section 3 introduces a dynamic adsorption-based DACCS model to assess the economic and climate benefits of a flexible DACCS system. Additionally, the optimization framework, the case study, and the examined scenarios are described. The scenarios differ in the input time series for electricity prices and the GHG emissions associated with electricity use, as well as in the capital expenditure of the adsorption-based DACCS system. Section 4 presents our results, evaluating the performance of flexible operation of adsorption-based DACCS across all scenarios. Finally, we draw our conclusions in Section 5.

2 Enhancing the flexibility of DACCS

The DSM methodology recognizes several methods of demand shaping, such as peak clipping, valley filling, and load shifting [18, 29]. The ability to shape the demand effectively depends on the flexibility of the process. For adsorption-based DACCS, demand flexibility can be provided through two approaches: (i) Incorporating energy storage units like batteries or heat storage, which tends to increase costs and complexity, or (ii) modifying the inherent flexibility of the process. Adsorption-based DACCS is particularly suited for modifying the process due to its wide Pareto-optimal operational range [14].

This chapter delves into how the standard cycles of adsorption-based DACCS can be modified to enhance flexibility for DSM without using energy storage. The study focuses on adapting the electricity demand of DACCS to respond flexibly to changes in the electricity prices and GHG emissions from electricity supply, aiming to reduce the overall net carbon removal costs. Note DACCS is used as a synonym for adsorption-based DACCS using amine-functionalized sorbents since this paper deals exclusively with this DACCS technology.

2.1 Adaptations of the standard cycles

The standard cycles used for amine-functionalized sorbents in DACCS applications are temperature vacuum swing adsorption (TVSA) and steam-assisted

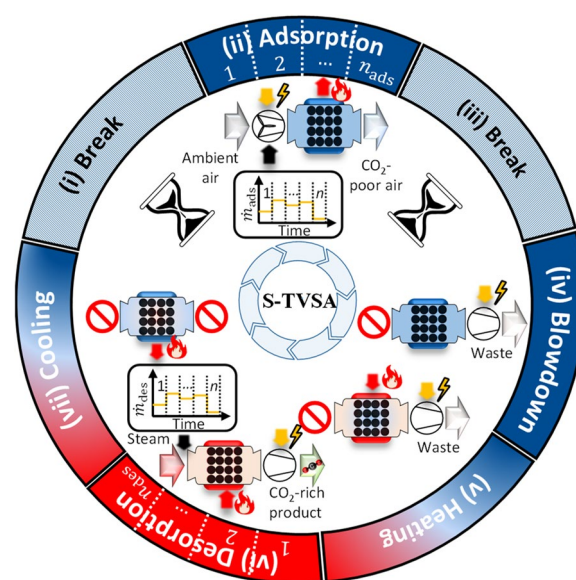


Fig. 1 Illustration of the proposed 7-phase steam-assisted temperature vacuum swing adsorption cycle, adapted to enhance flexibility. The enhanced cycle has the main phases (ii) adsorption, (iv) blowdown, (v) heating, (vi) desorption, (vii) cooling, and two breaks before (i) and after adsorption (iii). The adsorption and desorption phases are divided into sub-phases. For each sub-phase, different mass flows are possible (i.e., air mass flow during adsorption and steam mass flow during desorption). The color code of the outer ring represents the temperature trajectory in the DAC adsorption column: From ambient temperature (blue) to desorption temperature (red)

temperature vacuum swing adsorption (S-TVSA). Both cycles have five phases [30, 31]: (i) Adsorption, (ii) blowdown, (iii) heating, (iv) desorption, and (v) cooling. During desorption, the S-TVSA uses a steam mass flow to enhance desorption, while the TVSA does not use any steam. Detailed descriptions of all 5 phases are extensively covered in the literature [14, 30–32].

The 5-phase TVSA and 5-phase S-TVSA offer a wide operational range, which can be set by adjusting the process parameters, such as phase durations. Whereby, an inherent trade-off exists between efficiency and productivity (captured mass of CO₂ per time), leading to a Pareto frontier between the most efficient and most productive processes [14]. This trade-off offers intrinsic flexibility, as changing the process parameters changes the energy demand and the productivity.

To further increase the flexibility, we recommend extending the standard cycles by two additional phases and dividing the adsorption and desorption phases into several sub-phases (see Fig. 1). The two additional phases are breaks before and after the adsorption phase, where no activity occurs. These breaks can be strategically utilized to pause the entire operation. Pausing operation

further enhances flexibility and can be economically and ecologically beneficial in times of high electricity prices or large GHG emissions from electricity supply. Furthermore, breaks are crucial if the operation is tied to a non-permanent power supply, for example, when using curtailed wind energy [28]. Although breaks in or between the phases (iv) blowdown to (vii) cooling could further increase flexibility, the pressure of the adsorption columns is below ambient pressure, and the adsorption column may not be perfectly sealed. Hence, air would leak into the column, reducing the CO₂ purity and requiring more energy for the vacuum pump to extract the air again. Breaks are even more harmful between the phases (v) heating and (vii) cooling due to heat losses. Minimizing these heat losses would require an enormous design effort, which seems unjustified compared to the slightly increased flexibility. In contrast, the breaks before and after the adsorption phases do not negatively impact the energy demand. However, no CO₂ is captured during these breaks, which influences productivity and, thus, economic performance. Therefore, the use and duration of these breaks before and after the adsorption phase result in a trade-off between operating and investment costs.

In addition to these two new phases, the adsorption and desorption phases are divided into sub-phases (n_{ads} and n_{des} respectively). This subdivision allows independent air or steam mass flows during each sub-phase. As these mass flows are directly correlated to the energy demand and productivity of the DACCS system, segmenting these phases enhances flexibility. For instance, a high air mass flow during adsorption creates higher velocities within the column, which increases the pressure drop and fan's energy demand. Simultaneously, the high air mass flow accelerates adsorption, thereby increasing productivity. Similar correlations apply to the steam mass flow during desorption. A higher steam mass flow increases the energy demand, as additional energy is required to heat, vaporize, and superheat the water. On the contrary, the superheated steam rapidly heats the sorbent and efficiently displaces the desorbed CO₂. Consequently, desorption is faster, enhancing productivity but increasing energy demand [14].

Adapting the air and steam mass flows results in a trade-off between energy demand and productivity. This trade-off leads to varying optimal mass flows depending on fluctuating electricity prices or the electricity's GHG emissions. When more sub-phases are used, the mass flows can be adapted better to input changes. The number of sub-phase can vary from cycle to cycle. The number of sub-phases is selected to align with periods of constant electricity prices and GHG

emissions. Thus, the number of sub-phases depends on the phase times of adsorption and desorption and the frequency of changes in electricity prices and GHG emissions. From a technical perspective, the air mass flow could be adjusted by controlling the fan speed. A simple total evaporator could be combined with a controllable water pump to provide variable steam mass flows.

The proposed 7-phase cycle has a broad spectrum of adjustable process parameters (p_c) that enable flexible operation. These parameters include the phase times of the pre-adsorption break ($\tau_{\text{break,pre}}$), adsorption phase (τ_{ads}), post-adsorption break ($\tau_{\text{break,post}}$), and desorption phase (τ_{des}). Additional process parameters are the air mass flows in each adsorption sub-phase ($\dot{m}_{\text{ads},j}$) and the steam mass flows in each desorption sub-phase ($\dot{m}_{\text{des},j}$). The desorption pressure (p_{des}) is another vital process parameter and refers to the column pressure during phases (iv) to (vii). This extensive control via the process parameters significantly boosts the system's flexible operation. Overall, the proposed changes greatly increase the flexibility, making the presented 7-phase cycle practical for DSM.

3 Modelling and optimization of a flexible DACCS system

3.1 Dynamic DACCS system model

System-specific performance indicators are required to holistically evaluate a flexible operation of DACCS. A detailed dynamic DACCS system model is used to calculate these performance indicators. Figure 2 illustrates the DACCS system model, its components, and its interfaces. The DACCS system model is developed in *Modelica*, using components from our open-source sorption system library *SorpLib* [33], which is freely available on GitLab [34]. The parameterized model of the DACCS system described in this publication will be provided in a future version of *SorpLib*.

The DACCS system model is adapted from our previous publication [14]. A detailed description of all main components, their model equations, and main assumptions is provided in the Supporting Information and our previous publication [14]. Each component is briefly described below, with a focus on the most important aspects and the adaptations made.

- (i) The DAC plant is the main component of the DACCS system model. In the DAC plant, energy (mechanical and thermal) is needed to capture CO₂ from the air and to produce high-purity CO₂ at approximately one bar. The DAC plant model includes an dynamic adsorption column model [14], which periodically runs the 7-phase cycle described in Section 2. The dynamic adsorption

column model is 1D discretized in the flow direction using a finite volume approach. The dynamic adsorption column model is parameterized according to the model by Stampi-Bombelli et al. [30], using equilibrium and kinetic data of amine-functionalized cellulose (APDES-NFC) [35–37]. The performance of a single column is linearly scaled up to a plant capacity of 4 kt yr^{-1} to emulate the performance of Climeworks' Orca DAC plant [14, 38]. Additionally, the DAC plant accounts for the life-cycle GHG emissions of plant construction, sorbent production, and sorbent consumption based on the life-cycle assessment from Deutz and Bardow [13]. Finally, the DAC plant requires capital expenditure (CapEx) for construction, which varies greatly between different studies [11, 27, 28, 39]. Considering the uncertainty in the capital expenditure and the expectation that capital expenditure decreases with increasing installed capture capacity [31], we vary the capital expenditure for the considered scenarios (Section 3.3.2).

- (ii) Recooling dissipates heat from the process into the environment. A simple model of a dry cooler is employed, which has a specific power demand for a fan per discharged heat [40].

- (iii) The storage model includes the power demand of CO_2 compression for injection into the geological storage site. It is assumed that the storage is on-site, eliminating energy demand for transportation. Additionally, the storage process is considered leak-free, ensuring that all captured CO_2 is permanently stored without any losses.
- (iv) The energy supply includes a high-temperature heat pump (HTHP) that converts electricity into heat, providing all required process heat for the DACCS system. By using a high-temperature heat pump for heat supply, the DACCS model requires only electricity for operation. Electricity is modeled as grid electricity that is characterized by time-resolved prices ($c_{el}(t)$) and time-resolved specific GHG emissions attributable to the electricity production ($cc_{el}(t)$). These time-resolved inputs are critical to our model as they significantly influence the operating costs and environmental impact of the DACCS system. By incorporating these inputs, our model can simulate the economic and environmental performance of the DACCS system for various scenarios. The choice of input time series and the definition of different scenarios are explained in detail in 3.3.3 section.

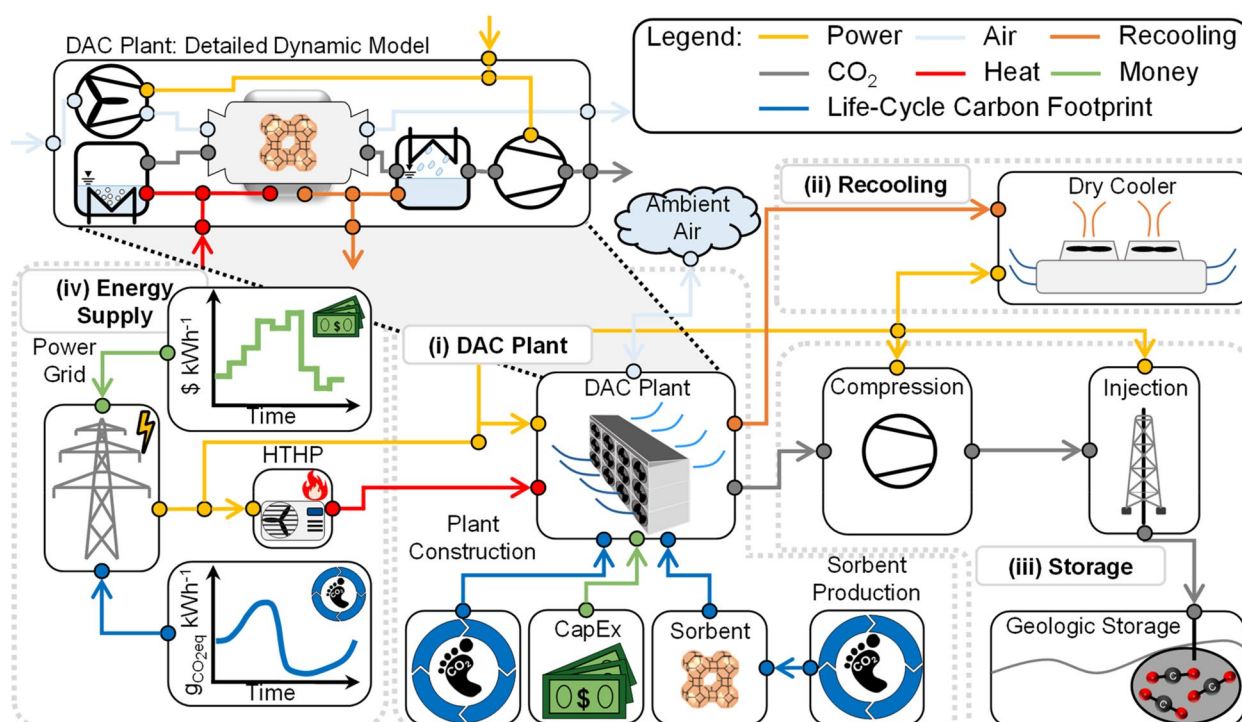


Fig. 2 Scheme of the DACCS system model, including the main components (i) DAC plant, (ii) recooling, (iii) storage and (iv) energy supply. Electricity flows are yellow and heat flows are red. Life-cycle GHG emissions are blue, CO_2 mass flows are grey, and economic flows are green. The scheme is adapted from Postweiler et al. [14] under CC BY 3.0 license

3.2 Performance evaluation

Sound performance indicators are essential for evaluating and optimizing DACCS systems [14]. The carbon removal efficiency (CRE) is a crucial indicator from a "techno-ecological" point of view [13, 15]:

$$CRE_i = \frac{m_{\text{CO}_2, \text{cap}, i} - CC_i}{m_{\text{CO}_2, \text{cap}, i}} = \frac{m_{\text{CO}_2, \text{net}, i}}{m_{\text{CO}_2, \text{cap}, i}}. \quad (1)$$

The carbon removal efficiency is calculated using the climate change impact (CC_i) from all life-cycle GHG emissions in the system and the total amount of captured CO_2 from air within one cycle ($m_{\text{CO}_2, \text{cap}, i}$). The climate change impact CC_i includes the allocated emissions from the electricity demand, the construction and end-of-life of the DACCS system, the sorbent consumption, and any potential leakages. The precise composition and calculation of the climate change impact CC_i are explained in detail in the Supporting Information. In the numerator of Eq. 1, the difference $m_{\text{CO}_2, \text{cap}, i} - CC_i$ is the net mass of removed CO_2 ($m_{\text{CO}_2, \text{net}, i}$). Consequently, the carbon removal efficiency is the ratio of the net to the gross mass of removed CO_2 .

In addition to a "techno-ecological" assessment, an economic assessment is also important. The gross carbon removal cost (GCRC) is suitable for determining the leveled cost of capturing CO_2 and is defined as follows [21]:

$$GCRC_i = \frac{C_i}{m_{\text{CO}_2, \text{cap}, i}}. \quad (2)$$

The total cost C_i is the sum of capital expenditure (CapEx) on the one hand and operational expenditure (OpEx) on the other:

$$\begin{aligned} C_i &= \text{CapEx}_i + \text{OpEx}_i \\ &= \text{CapEx}_{\text{DACCS}} AF \frac{\tau_{\text{cycle}, i}}{\tau_{\text{year}}} + \text{OpEx}_{\text{DACCS}, i}. \end{aligned} \quad (3)$$

The total capital expenditure of the DACCS system ($\text{CapEx}_{\text{DACCS}}$) is allocated to one cycle by using the annuity factor (AF), the cycle duration ($\tau_{\text{cycle}, i}$), and the number of hours per year (τ_{year}). For the operational expenditure ($\text{OpEx}_{\text{DACCS}, i}$), we assume that only the costs of electricity supply are significant, while all other operational expenditures are negligible. A detailed breakdown of the total cost for one cycle i is provided in the Supporting Information.

Combining the carbon removal efficiency and gross carbon removal cost forms the net carbon removal cost (NCRC), which integrates both the "techno-ecological" and economic dimensions [7, 39]:

$$\begin{aligned} \text{NCRC}_i(p_c, cc_{\text{el}}(t), c_{\text{el}}(t)) &= \frac{\text{GCRC}_i(p_c, cc_{\text{el}}(t), c_{\text{el}}(t))}{\text{CRE}_i(p_c, cc_{\text{el}}(t), c_{\text{el}}(t))} \\ &= \frac{C_i}{m_{\text{CO}_2, \text{net}, i}}. \end{aligned} \quad (4)$$

The net carbon removal cost gives a price to the actual benefit of DACCS, i.e., the net removed CO_2 mass from the atmosphere. Hence, the net carbon removal cost is used to evaluate the performance of the flexible DACCS system. The difference between net and gross carbon removal costs is called emission excess. Emission excess refers to the cost increase due to the life-cycle GHG emissions of the DACCS system.

3.3 Case study and optimization framework

3.3.1 Optimization framework

The DACCS process can be designed for two cases: (i) Steady-state operation or (ii) flexible operation. The electricity prices (C_{el}) and GHG emissions of electricity supply (CC_{el}) are assumed to be constant for steady-state operation. Then, the DACCS process is optimized by minimizing the net carbon removal costs (NCRC) for a single cycle, which is repeated over the considered period of one week:

$$\begin{aligned} \min_{p_c} \quad & \frac{C_i(x(t), z(t), cc_{\text{el}}, c_{\text{el}}, p_c)}{m_{\text{CO}_2, \text{net}, i}(x(t), z(t), cc_{\text{el}}, c_{\text{el}}, p_c)} && \text{objective functions} \\ \text{s.t.} \quad & \dot{x} = f(x(\tau), z(\tau), cc_{\text{el}}, c_{\text{el}}, p_c) && \text{dynamic model} \\ & 0 = g(x(\tau), z(\tau), cc_{\text{el}}, c_{\text{el}}, p_c) && \\ & x(t=0) = x(t=\tau_{\text{cycle}}) && \text{cyclic steady - state.} \end{aligned} \quad (5)$$

The optimization is solved subject to the differential-algebraic system equations of the dynamic DACCS model and a cyclic steady-state condition. The differential states are denoted as x , algebraic states as z , and the degrees of freedom in the optimization are the process parameters (p_c). The process parameters include the mass flows ($\dot{m}_{\text{ads}, i}$) during the adsorption phase, the phase times ($\tau_{\text{break}, \text{pre}}, \tau_{\text{ads}}, \tau_{\text{heat}}, \tau_{\text{des}}, \tau_{\text{break}, \text{post}}$), and the desorption conditions ($p_{\text{des}}, \dot{m}_{\text{des}, j}$). Table 1 lists all process parameters of the optimization and their upper and lower bounds. Note that the phase times of the blow-down (τ_{blow}) and heating phase (τ_{heat}) as well as the desorption temperature (T_{des}) are fixed because our previous study has shown that these process parameters have limited impact on the performance of the DACCS system [14]. For example, the desorption temperature is almost always set to its upper bound of 100 °C [14]. Therefore, these parameters were excluded from the process optimization to reduce computational effort.

Steady-state operation is not applicable when evaluating flexible DACCS operation with varying prices and GHG emissions from electricity supply. In addition to the resulting changing cycles, the start values of each cycle

Table 1 Process parameters with their lower and upper bounds for process optimization.^aNew boundaries for the “Restricted Flexible cycle” (cf. Section 4.5).^bMass flows refer to the mass flows of one column

Process parameters	Lower bound	Upper bound	
$\tau_{\text{break,pre}}$	1	28,800	s
τ_{ads}	3,000 8,280 ^a	21,600	s
$\tau_{\text{break,post}}$	1	28,800	
$\tau_{\text{blow (fixed)}}$	30	30	s
$\tau_{\text{heat (fixed)}}$	64	64	s
τ_{des}	5,000 8,280 ^a	21,600	s
$\dot{m}_{\text{ads},i}^b$	1e-5	2e-4	$\frac{\text{kg}}{\text{s}}$
$\dot{m}_{\text{des},j}^b$	1e-9	1e-7	$\frac{\text{kg}}{\text{s}}$
p_{des}	0.01	0.3	bar
$T_{\text{des(fixed)}}$	100	100	°C

i depend on the end values of the previous cycle ($i - 1$), demonstrating that all cycles are interconnected for flexible operation. One approach is to optimize all cycles within a given time period simultaneously. However, this approach is not feasible due to the complexity of the DACCS system model: In flexible operation, each cycle can be different, and each cycle has 17 process parameters. Thus, the total number of degrees of freedom accumulates quickly for multiple cycles. Considering one week which typically includes 20 cycles, the optimization problem has 340 degrees of freedom. Hence, solving the optimization problem rigorously is not a viable option.

To manage the computational effort for long periods, we use a rolling horizon algorithm for process optimization [41, 42]. In a rolling horizon algorithm, the total time period is divided into shorter time intervals, with the length of each interval varying based on the application. For the DACCS application, each time interval corresponds to the length of one cycle. Figure 3 presents a schematic depiction of the rolling horizon algorithm employed for the DACCS application.

For each step k of the rolling horizon algorithm, a process optimization is carried out for the time intervals k (current step, cf. Figure 3) and all intervals up to $k + n_{\text{hl}}$ (foresight, cf. Figure 3). Herein, n_{hl} is the horizon length and defines the number of additional time intervals considered per step, providing foresight into upcoming intervals. Having optimized the time intervals k up to $k + n_{\text{hl}}$, the results for interval k are fixed, while the results for all intervals beyond k are discarded. The discarded time intervals have been considered to account for effects of the subsequent intervals on interval k . The rolling horizon algorithm runs through the entire

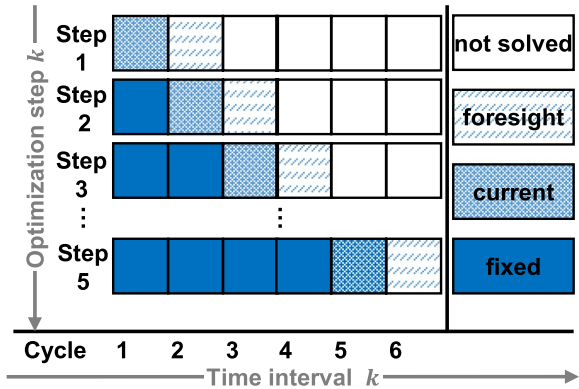


Fig. 3 Schematic depiction of the rolling horizon algorithm with a horizon length n_{hl} of one (adapted from Glomb et al. [43]). Each time interval corresponds to a single cycle and the horizon length defines how many upcoming intervals (“foresight”) will be optimized alongside the current interval k (“current”). Once step k is optimized, the solution for the “current” interval is stored (hence “fixed”) for future use, while the solutions for the foresight intervals are discarded. Consequently, “fixed” indicates intervals whose results are retained, “current” is the interval being optimized in a given step, and “foresight” covers intervals beyond step k that are also solved but only to guide the optimal solution for the current step

period consecutively, with each step k using (i) all start data from previous cycles and (ii) input data (e.g., electricity prices and specific GHG emissions of electricity) for intervals k to $k + n_{\text{hl}}$. To account for all previous data, the objective function includes the total cost and total net removed mass of CO_2 from all previous steps (i.e., $\sum_{i=0}^{k-1} C_i$ and $\sum_{i=0}^{k-1} m_{\text{CO}_2,\text{net},i}$ respectively). Additionally, in step k , the dynamic model is initialized with the differential states from the end of step $k - 1$, ensuring quasi-continuous operation over all time intervals. The optimization problem for each step k of the rolling horizon algorithm is formulated as follows:

$$\begin{aligned}
 & \min_{\mathbf{p}_{\text{c},k} \dots \mathbf{p}_{\text{c},k+n_{\text{hl}}}} \frac{\sum_{i=0}^{k-1} C_i + \sum_{i=k}^{k+n_{\text{hl}}} C_i(x(t), z(t), c_{\text{el}}(t), p_{\text{c}})}{\sum_{i=0}^{k-1} m_{\text{CO}_2,\text{net},i} + \sum_{i=k}^{k+n_{\text{hl}}} m_{\text{CO}_2,\text{net},i}(x(t), z(t), c_{\text{el}}(t), p_{\text{c}})} && \text{objective function} \\
 & \text{s.t.} && \\
 & \dot{\mathbf{x}} = f(\mathbf{x}(\cdot), \mathbf{z}(\cdot), \mathbf{p}_{\text{c}}, c_{\text{el}}, c_{\text{el}}) && \text{dynamic model} \\
 & 0 = g(\mathbf{x}(\cdot), \mathbf{z}(\cdot), \mathbf{p}_{\text{c}}, c_{\text{el}}, c_{\text{el}}) && \\
 & \mathbf{x}_k(t = \tau_{\text{start},k}) = \mathbf{x}_{k-1}(t = \tau_{\text{end},k-1}) && \text{initialization}
 \end{aligned} \tag{6}$$

This rolling horizon algorithm minimizes the average net removal costs from the beginning to the end of the $k + n_{\text{hl}}$ intervals. Once the horizon length encompasses the number of all cycles in the time period, optimization of the entire time period is achieved at once.

Reducing horizon length n_{hl} degrades the solution quality to some extent but simplifies the optimization problem, making it solvable with less computing effort. A horizon length n_{hl} of one was selected, resulting in 34 degrees of freedom. To solve the optimization problem, the dynamic DACCS model from *Modelica* is

embedded into an optimization framework in Python using the Functional Mock-up Interface [44]. The optimization framework is based on our implementation of the rolling horizon algorithm and utilizes a heuristic particle swarm optimization algorithm from the Python package *Pymoo* [45, 46].

The rolling horizon algorithm offers the significant advantage of avoiding unrealistically long ideal foresight. The optimization of each upcoming cycle is based solely on the in-puts available for the next few hours, which is exactly what a real-world online process optimization is limited to. For instance, electricity prices are foreseeable via the day-ahead or intraday markets, providing the data required for online process optimization. By combining these short-term forecasts with a digital twin of a DACCS system, real-world online process optimization becomes feasible. The rolling horizon algorithm imitates this limited foresight of real-world online process optimization. Consequently, rolling horizon algorithms are particularly valuable for providing realistic estimates of potential improvements in real-world applications.

3.3.2 Investigated case study

Our analysis includes a variety of scenarios, each with unique characteristics. All scenarios have the same fundamental parameters including environmental conditions like temperature and relative humidity, and the DACCS system configuration (Fig. 2). Ambient conditions, such as temperature and relative humidity, crucially influence the performance of DACCS systems [27, 47, 48]. Annual average values for Aachen, Germany, were used for all calculations, as varying ambient conditions are beyond the scope of this study (Table 2). However, our modeling framework is also capable to incorporate varying ambient conditions. The DACCS system is designed for a nominal

annual productivity of 4 kt yr^{-1} . The nominal productivity is reached using a reference process, but the DACCS system also allows for Pareto-optimal operation (productivity vs. carbon removal rate) within a wide range from approximately 1.5 to 4.5 kt yr^{-1} by tweaking process parameters. For comprehensive details on this scaling, the reference processes, and the Pareto-optimal operation, refer to our previous publication [14]. Further, the DACCS plant is assumed to operate over a lifespan of 20 years, and a 7 % interest rate is applied to annualize capital expenditure [21]. Table 2 summarizes all case study specific parameters.

3.3.3 Investigated scenarios

We categorize our scenarios into two dimensions: (i) The time series used as inputs, namely the prices c_{el} and specific GHG emissions cc_{el} of electricity supply, and (ii) the capital expenditure of the DACCS system. For the time series inputs, either historical data or forecasts can be employed, each presenting distinct benefits and drawbacks.

Historical data concerning electricity prices and specific GHG emissions of electricity supply are readily accessible and reliable. However, historical data depend on existing configurations of energy systems, which might impose limitations for analyses for the future. For instance, the current electrical system in Germany is characterized by high specific GHG emissions, thus rendering the implementation of grid-based DACCS nearly untenable [13, 14, 47]. Regarding forecast data, Germany has ambitious decarbonization targets that largely depend on intermittent renewable sources like wind and solar. Thus, grid-based DACCS could become viable in Germany's future electrical system. Further, Germany's future electrical power systems seem to be a suitable case

Table 2 Fundamental parameters and input data of the case study and the different scenarios. The combinations of different CapEx and input data examined are marked with an X for the scenarios

Case Study					
Ambient temperature	10.1	°C	[49]		
Relative humidity	77	%	[49]		
Ambient pressure	1.01325	bar	[49]		
Mole fraction of CO ₂ in dry air	407	ppm	-		
Nominal annual productivity	4	kt yr ⁻¹	[14]		
DACCS system lifespan	20	yr	[21]		
Interest rate	7	%	[21]		
Scenarios					
Input Data (Forecast Germany's electric power system)					
CapEx		Summer 2030	Winter 2030	Summer 2040	Winter 2040
Low	3 € mio.	X	-	-	-
Medium	7 € mio.	X	X	X	X
High	11 € mio.	X	-	-	-

study for the flexible operation due to the anticipated high shares of intermittent renewable energy sources.

Currently, some of the authors are developing a forecasting model for energy system transitions [50]. This model is based on *SecMOD*, which is a framework for the optimization and life-cycle assessment of multi-sector energy systems [51]. The model is based on data of the German electric power system in 2020. Using the model, a transition pathway is outlined of how the system evolves over time. This pathway involves optimizing the transition to meet future GHG emission targets while minimizing energy costs. Upon defining the targeted future electric power system, time series data can be generated for electricity prices and specific GHG emissions of electricity supply using comprehensive weather data and demand profiles. For better readability, we use the term "electricity prices," actually referring to the electricity market clearing prices.

This study employs a forecast framework to project Germany's future electric power system for 2030 and 2040. For both years, we exemplarily generated time series data for seven days in January (13th–20th) and seven days in July (13th–20th). During these periods, prices and specific GHG emissions of electricity supply were calculated on an hourly basis. This process yielded four scenarios based on the input time series: Summer-2030 (*cf.* Figure 5), Winter-2030 (*cf.* Figure 8), Summer-2040, and Winter-2040. Detailed results for all four scenarios are presented in the Supporting Information.

During the summer weeks, daily fluctuations in prices and specific GHG emissions of electricity are distinctly noticeable, primarily due to photovoltaic-driven electricity generation (Fig. 5a). In contrast, these daily fluctuations are less pronounced during the winter weeks. Instead, more substantial and prolonged variations are observed, primarily driven by the significant contribution of wind power (Fig. 8a). The disparity between the seasons highlights the dynamic nature of renewable energy integration.

Moving from 2030 to 2040, the fluctuations in electricity prices and specific GHG emissions of electricity supply intensify significantly. In 2040, surplus electricity drives the electricity prices down to 0 € MWh⁻¹ in some winter and summer periods. In other periods, renewable sources are insufficient to meet demand, resulting in electricity prices being significantly increased by the reliance on costly backup power plants, such as gas-fired facilities. These high fluctuations underscore the challenges for a transition to a renewable-based energy system, emphasizing the potential of flexible DACCS operations to counteract these fluctuations.

The forecast model assumes that power plants are dispatched depending on their marginal generation cost. However, it is unclear if future electricity markets

will operate this way. Additionally, this simplified forecast model cannot reflect specific market behavior, such as negative electricity prices. Hence, our forecast model may not perfectly predict the future of Germany's electricity system, nor do we claim that selecting specific weeks in January and July provides a comprehensive overview of the energy system throughout the year. However, this work does not aim to present an exhaustive forecast of Germany's electric energy system but rather to show the potential of flexible DACCS operation within an electric energy system characterized by a high share of renewables.

In addition to the four scenarios for the time series inputs, scenarios for different capital expenditures associated with a 4 kt yr⁻¹ DACCS system are also explored. The literature reports significant variability in capital expenditure estimates for such DACCS systems [11, 21, 27, 39, 52]. For example, the Climework's Orca plant has a productivity of 4 kt yr⁻¹ and its capital expenditures are reported from 10–15 \$ million (equivalent to roughly 9–14 € million) [52]. Conversely, Fasihi et al. report a capital expenditure of approximately 2.9 € million for a DACCS system with comparable technology and productivity in their techno-economic assessment [11]. Considering the anticipated decrease in capital expenditure with extending productivity [39], we assume that all future DACCS plants should at least match the economics of the Climework's Orca plant. Accordingly, 11 € million is used as our High-CapEx scenario, 7 € million as our base case (Medium-CapEx) and 3 € million as our Low-CapEx scenario. These three scenarios enable us to assess the DSM potential of DACCS under various assumptions about capital expenditures for a DACCS system with productivity of 4 kt yr⁻¹. Intuitively, high capital expenditures would lead to maximum productivity, whereas low capital expenditure give more weight to the operational expenditures and thus, would render flexible DACCS operation for DSM more attractive.

Overall, the combination of different time series for inputs and varying capital expenditures results in six scenarios (*cf.* Table 2): three for analyzing the influence of capital expenditures (Summer-2030 with High-, Medium-, and Low-CapEx) and three for examining the influence of input data (Winter-2030, Summer-2040, and Winter-2040, all with Medium-CapEx).

4 Results

In this chapter, we compare the performance of flexible DACCS operation with steady-state operation. For flexible operation, we use the rolling horizon algorithm (Eq. 6). For steady-state operation, we optimize the DACCS process for one cycle using average inputs for

the prices and GHG emissions of electricity (Eq. 5). Two approaches are employed when averaging the input data. The "Annual cycle" uses the mean values of the inputs for one year. Since comprehensive year-round data is unavailable, these values are approximated using averages from the selected summer and winter weeks. The "Seasonal cycle" uses weekly average values for the specific weeks in summer or winter. Thus, the "Seasonal cycle" introduces a form of flexible operation, as the DACCS system does not operate in a complete steady-state but rather uses optimized process parameters for different seasons.

4.1 The "Annual Cycle"

Before delving into the flexible operation, the "Annual cycle" is examined. For the "Annual cycle", we employ our developed 7-phase cycle (cf. Section 2). The 7-phase cycle is designed for flexible operation, which limits its advantages in steady-state operation. For instance, the break phases are always set to zero for optimal steady-state operation, as the potential benefits of these breaks—such as waiting for "cheaper" or "cleaner" electricity—cannot be realized. Still, the 7-phase cycle does offer advantages over the 5-phase cycle in steady-state operations. Schellevis has already highlighted benefits of variable air mass flow during adsorption and a variable steam mass flow during desorption [53]. At the beginning of the adsorption phase, the CO₂ adsorption rate is primarily limited by the concentration of CO₂ in the gas phase, making a high air mass flow particularly useful. At the end of the adsorption phase, the limitation is more due to adsorption kinetics rather than CO₂ concentration, meaning a high air mass flow does not significantly enhance the adsorption rate but increases energy demand. Therefore the 7-phase cycle is expected to have advantages over the standard 5-phase cycle.

The 7-phase and 5-phase "Annual cycle" have roughly the same specific heat demands defined as the total heat demand of the DAC plant per captured mass of CO₂ (cf. Figure 2), with 3.5 MWh_{th} t_{CO₂,cap}⁻¹ for both cycles. The specific electricity demand, which accounts for the total electricity consumption of the DAC plant, recooling, and storage per captured mass of CO₂ (cf. Figure 2), differs slightly between the two cycles, with values of 0.78 MWh_{el} t_{CO₂,cap}⁻¹ for the 7-phase and 0.84 MWh_{el} t_{CO₂,cap}⁻¹ for the 5-phase cycle. Among different DACCS processes, the two most technical advanced ones — adsorption-based and absorption-based — exhibit specific heat demands ranging from 0.9 to 13.4 MWh_{th} t_{CO₂,cap}⁻¹ and specific electricity demands between 0.1 and 0.8 MWh_{el} t_{CO₂,cap}⁻¹ [9, 54]. More importantly, the heat and electricity demand reported for a real-world DACCS system of a similar design is reported

as 3.3 MWh_{th} t_{CO₂,cap}⁻¹ and 0.7 MWh_{el} t_{CO₂,cap}⁻¹, respectively [13]. Despite minor discrepancies in the precise heat and power demand, the simulation is highly congruent with the actual values reported for a real-world DACCS systems of similar design. All process parameters and an analysis of the heat and electricity demand of the 7-phase and 5-phase "Annual cycle" is given in the Supporting Information. For a detailed thermodynamic analysis of the employed DACCS system model, including specific exergy demand, carbon removal efficiency, and an evaluation of process parameters for Pareto-optimal operation, we refer readers to our previous open-access publication [14].

Figure 4a shows the net carbon removal costs for 2030 for the 5-phase and 7-phase cycle. The 7-phase cycle (388.6 € t_{CO₂}⁻¹) exhibits 9.8 € t_{CO₂}⁻¹ (-2.5 %) lower net carbon removal costs than the 5-phase cycle (398.4 € t_{CO₂}⁻¹). The capital expenditure allocation of both cycles is almost identical (see Fig. 4a), which is reasonable since both cycles have nearly the same cycle time and the capital expenditure allocation depends primarily on the cycle length. The operational expenditures attributed to the HTHP (i.e., heat supply) are also nearly identical for both cycles. A key difference is the operational expenditure attributed to the fan, where costs can be reduced from approximately 9.7 € t_{CO₂}⁻¹ to 7.9 € t_{CO₂}⁻¹. The costs for compression and cooling remain nearly constant between the two cycles. The reduction in emission excess of 7.1 € t_{CO₂}⁻¹ (-4.5 %) is the largest single reduction between both cycles (emission excess: 5-phase cycle 158.41 € t_{CO₂}⁻¹ and 7-phase cycle 151.3 € t_{CO₂}⁻¹). For the emission excess, the indirect GHG emissions due to energy consumption are imperative. In the 7-phase cycle, the fan consumes less energy (-18.5 %), which directly results in the above mentioned fan's cost reductions of 1.8 € t_{CO₂}⁻¹ (-18.5 %). More importantly, the reduced energy demand of the fan leads to fewer indirect GHG emissions from electricity supply, thus reducing the emission excess (-4.5 %) and finally reducing the net carbon removal costs.

Figure 4b shows the inlet mass flow during adsorption and desorption scaled to their upper bounds to examine the reduction of the fan's energy demand in detail. While the 5-phase cycle has a continuous mass flow, the 7-phase cycle uses a stepwise declining mass flow. The stepwise decline occurs in 1-hour intervals because the number of sub-phases was aligned with the hourly intervals of the variable input data (prices and GHG emissions of electricity). We expect that even greater savings could be realized with a continuous or at least more granular mass flow function. However, such continuous optimal fan control is beyond the scope of this analysis and could be a subject for future research. For the desorption phase, Fig. 4b

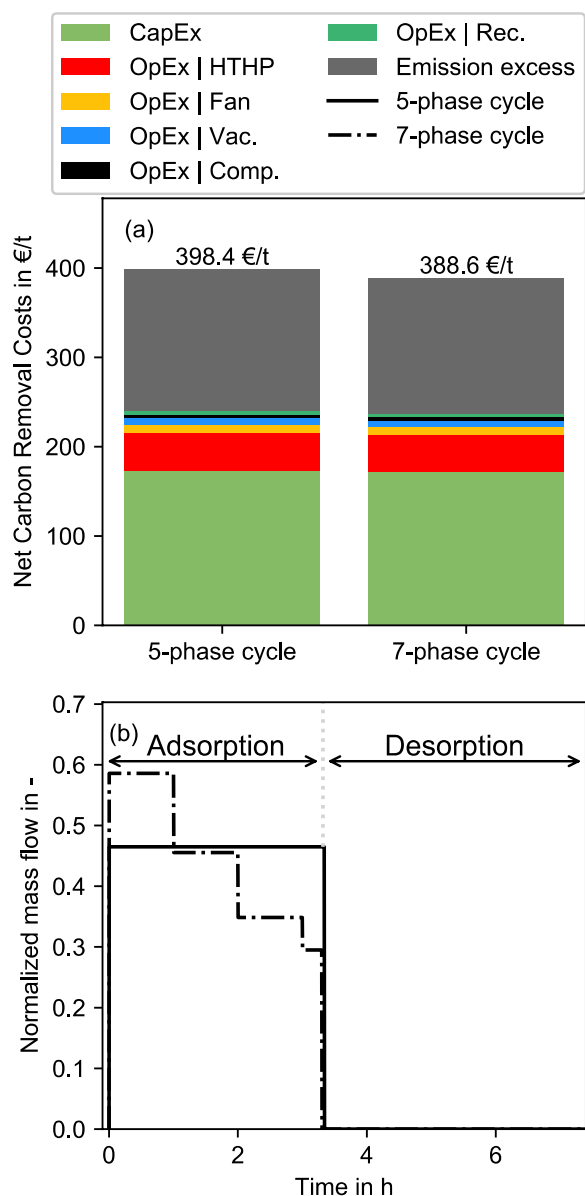


Fig. 4 Steady-state operation of the “Annual cycle” for 2030 comparing the standard 5-phase and the developed 7-phase cycle. **a** Illustrates the total net carbon removal costs (NCRC) and the breakdown of overall costs into capital expenditure (CapEx), operational expenditure (OpEx) for different parts of the DACCS system, and emission excess. **b** Displays the normalized mass flow at the entry of the DACCS system, highlighting the primary differences between the two cycles

shows that beyond 3.3 hours, the scaled inlet mass flow is zero. Thus, the optimal steam mass flow during desorption is zero, effectively leading the (S)-TVSA operating as a TVSA. This finding is consistent with results from our previous analysis, where the S-TVSA only outperforms the TVSA when using very “clean” electricity [14].

4.2 Flexible DACCS operation in summer 2030

In this chapter, flexible DACCS operation is analyzed and consistently compare the improvements to the steady-state cycle discussed in Section 4.1. First, we examine the Summer-2030 | Medium-CapEx scenario for flexible DACCS operation in detail.

Figure 5a shows the time-resolved inputs, i.e., electricity prices and GHG emissions of electricity supply. The electricity price varies between 8.2 € MWh⁻¹ and 63.1 € MWh⁻¹, while GHG emissions range from 49.5 to 341 kg MWh⁻¹. Figure 5a highlights that the electricity prices and GHG emissions are highly correlated, varying within a 24-hour interval between local maxima and minima. This 24-hour pattern is expected for an energy system with a significant share of solar electricity. The first operation mode examined is the “Annual cycle”, which now operates with time-resolved inputs instead of constant ones (cf. Section 4.1). Still, the “Annual cycle” runs a steady-state operation (i.e., all cycles are identical) but is now analyzed with time-resolved inputs for electricity prices and GHG emissions. Figure 5b shows the moving average of the net carbon removal costs for the “Annual cycle”. The moving average of the net carbon removal costs is calculated for each cycle, meaning that constant values can be attributed to one cycle.

In the first few cycles, there are strong fluctuations in the moving average of the net carbon removal costs, which diminish over time. The diminishing fluctuations are caused by the moving average: Initially, each individual cycle has a dominating impact, which decreases with time. For the entire week, the “Annual cycle” results in average net carbon removal costs of 528.2 € t_{CO₂}⁻¹. As a step towards flexible operation, Fig. 5b shows the “Seasonal cycle”. The “Seasonal cycle” is also a steady-state cycle, but its process parameters were specifically optimized for the average inputs of the week considered. The “Seasonal cycle” achieves net carbon removal costs of 514.3 € t_{CO₂}⁻¹. Thus, optimizing the process for specific periods can already achieve a 2.6 % reduction in average net carbon removal costs. The “Flexible cycle” further reduces the average net carbon removal costs to 488.1 € t_{CO₂}⁻¹ (Fig. 5b). Thus, flexible operation can reduce the net carbon removal costs by 7.6 % compared to the “Annual cycle” and by 5.1 % compared to the “Seasonal cycle” with the same DACCS system and the same time-resolved inputs.

To understand the significant reduction in average net carbon removal costs from flexible operation, some specific flexible cycles are analyzed in detail. During the period from day 0.4 to 0.8, electricity prices and GHG emissions are very low (Fig. 5a). To take full advantage of this “cheap” and “clean” electricity, the flexible operation uses two short cycles, whereas the steady-state “Annual

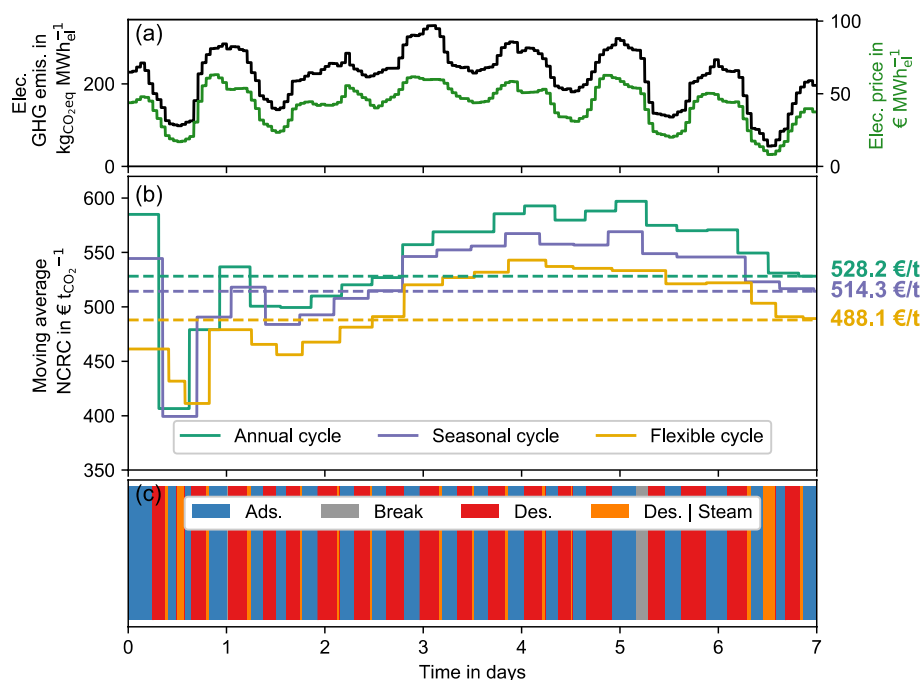


Fig. 5 Flexible DACCS operation for the Summer-2030 | Medium-CapEx scenario. **a** Inputs of the DACCS model: Electricity price in green and GHG emissions in black. **b** Moving average net carbon removal costs (NCRC) for three different operating modes: (i) the steady-state “Annual cycle”, (ii) the steady-state “Seasonal cycle”, and (iii) the “Flexible cycle.” The dashed lines represent the respective weekly average net carbon removal costs for each operating mode. **c** Phases of the “Flexible cycle,” providing insights into flexible operation strategies. Note that the phases blowdown, heating, desorption, and cooling are summarized as desorption

cycle” and “Seasonal cycle” only perform one cycle. Short cycles are generally more productive, meaning more CO₂ is captured per time. This higher productivity comes at the cost of efficiency, resulting in higher specific energy demands for short cycles [14]. When electricity is “cheap” and “clean,” the higher specific energy demand has less impact on the net carbon removal costs, and the benefits of higher productivity, i.e., lower specific capital expenditure, outweigh the disadvantage of increased specific energy demand.

To gain more insight into the flexible operation, Fig. 5c shows the individual phases of the different cycles. In the previously mentioned period from day 0.4 to 0.8, desorption is performed with a steam mass flow. Using a steam mass flow is reasonable and can be attributed to the described trade-off between productivity and efficiency. The steam mass flow increases the productivity of the cycle but at the cost of higher specific energy demand [14]. Again, the benefits of higher productivity outweigh the advantages of higher efficiency when electricity is “clean” and “cheap.”

The opposite effect can be observed when examining the period from day 0.8 to 1.3 in detail. This period is characterized by high electricity prices and high GHG emissions. During this period, the efficiency of the

process is more important than productivity, resulting in long cycles and almost no steam mass flow during desorption. However, steam is used at the end of desorption because electricity prices and GHG emissions are declining. Additionally, short steam usage towards the end of desorption is observed in many of the “Flexible cycles” (cf. Figure 5c). The steam mass flow during desorption has two main benefits: (i) Steam quickly heats the sorbent through direct heating. (ii) Steam reduces the partial pressure of CO₂, allowing more CO₂ to be desorbed. The faster heating favors steam flow at the beginning of desorption, which is only observed when electricity is “cheap” and “clean.” The reduced partial pressure is particularly effective at the end of desorption. Due to the slower adsorption kinetics of CO₂ compared to water, the gas phase at the end of desorption contains almost pure CO₂. Here, a small mass flow of steam can lower the partial pressure of CO₂, increasing desorption and productivity. Towards the end of desorption, the CO₂ mass flow is low. Thus, only small amounts of steam are needed to significantly reduce the partial pressure, keeping the higher energy demand for providing steam at the end of desorption low. Hence, the advantages of steam flow at the end of desorption are dominating in many cycles in the Summer-2030 | Medium-CapEx scenario.

When looking for breaks in the “Flexible cycle” (Fig. 5c), almost none are observed because the net carbon removal costs are dominated by capital expenditure in the Summer-2030 | Medium-CapEx scenario (cf. Figure 4a). In such cases, a break is more expensive than using slightly more “expensive” or “dirtier” electricity. Still, a 3-hour break exists at the beginning of day 5: The cycle is in adsorption during a peak period of electricity prices and GHG emissions. The beginning of desorption is the most energy-intensive part of the whole cycle as the adsorption column is heated, creating a peak in heat demand. To shift this peak to periods when electricity is “cheap” and “clean”, a break is introduced. This point in time is particularly suitable for a break as it corresponds to one of the steepest declines in electricity prices and GHG emissions in the entire time series.

In the next sections, we analyze the impact of capital expenditure and input data on net carbon removal costs.

4.3 The impact of capital expenditure

As shown in Fig. 4 for the steady-state operation, the capital expenditure of the DACCS system has the largest impact on the net carbon removal costs. Therefore, the impact of capital expenditure variations is examined. For this analysis, the summer 2030 input data is combined with three different capital expenditures resulting in the

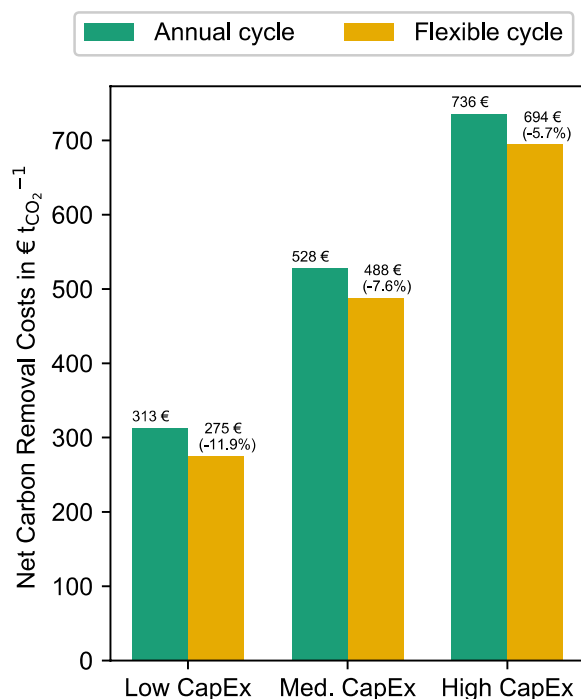


Fig. 6 Average net carbon removal costs for the Summer-2030 scenario comparing the “Annual cycle” and the “Flexible cycle” across three capital expenditure (CapEx) scenarios: Low (3 € million), Medium (7 € million), and High (11 € million)

three scenarios Summer-2030 | (i) Low-CapEx (3 € million), (ii) Medium-CapEx (7 € million), and (iii) High-CapEx (11 € million).

Figure 6 shows the average net carbon removal costs for the steady-state “Annual cycle” and “Flexible cycle”. The “Flexible cycle” achieves a reduction in net carbon removal costs of 11.9 % for the Low-CapEx, of 7.6 % for the Medium-CapEx, and of 5.7 % for the High-CapEx scenario.

The “Flexible cycle” tends to operate most efficiently for the Low-CapEx scenario: In direct comparison, Low-CapEx leads to 18 cycles, Medium-CapEx to 20 cycles, and High-CapEx to 21 cycles for one week. As longer cycles tend to be more efficient but less productive, productivity becomes more important than efficiency with increasing capital expenditure. In other words, as capital expenditure increases, the share of operational expenditure decreases, making productivity more important than efficiency.

Furthermore, the use of breaks also changes with the capital expenditure of the DACCS system. The Low-CapEx scenario has the same breaks as the Medium-CapEx scenario (cf. Figure 5c). In contrast, there are no breaks for the High-CapEx scenario since downtimes impact the net carbon removal costs more significantly, and waiting for “cleaner” and “cheaper” electricity is no longer economically viable. For the Low- and High-CapEx scenarios, a figure similar to Fig. 5 is included in the Supporting Information.

Overall, the “Flexible cycle” can reduce the net carbon removal costs between 5.7 % and 11.9 % for different capital expenditures of the DACCS system. Even for the High-CapEx scenario as a worst-case assumption for future DACCS systems, the net carbon removal costs can be reduced by 5.7 %.

4.4 The impact of electricity prices and GHG emissions

Figure 7 shows the average net carbon removal costs for the Winter-2030, Summer-2030, Winter-2040, and Summer-2040 scenarios all using Medium-CapEx. For the Summer-2040 and Winter-2040 scenarios, the net carbon removal costs can be greatly reduced by 20.1 % and 12.9 %, respectively, when comparing the “Annual cycle” with the “Flexible cycle”. These high reductions can primarily be explained by (i) the high average electricity price and (ii) its large fluctuations in 2040. For example, there are periods with very high electricity prices of around 340 € MWh⁻¹ in the Summer-2040 scenario, but also phases where the price drops to 0 € MWh⁻¹ due to an oversupply from renewables. With such large fluctuations in the electricity prices, flexible DACCS operation is particularly recommended to minimize the net carbon removal costs. Although the reduction potential is very high for the Summer-2040 and Winter-2040 scenarios,

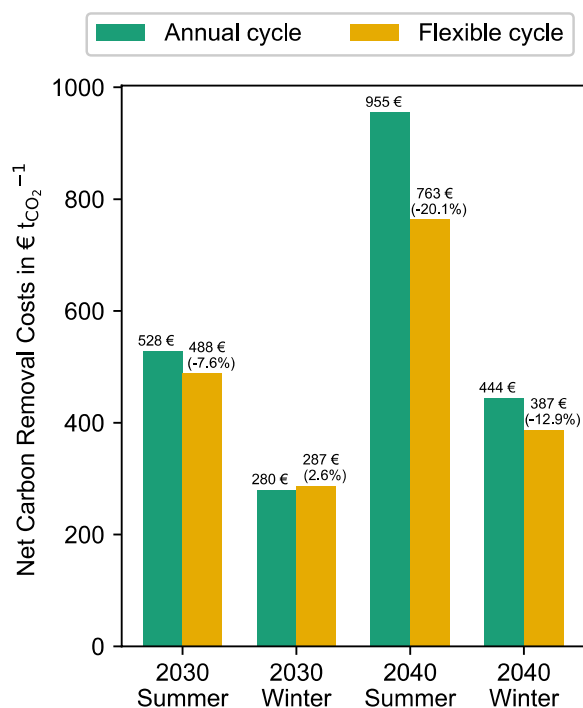


Fig. 7 Average net carbon removal costs for the Medium-CapEx scenario across different scenarios: Summer-2030, Winter-2030, Summer-2040, and Winter-2040

the absolute costs for flexible operation remain relatively high at 763 € t_{CO₂}⁻¹ and 387 € t_{CO₂}⁻¹, respectively. The absolute costs remain high due to the high average electricity price. A detailed illustration of the Summer-2040 and Winter-2040 scenarios can be found in the Supporting Information.

The Summer-2030 scenario has already been discussed in detail in Section 4.2. Interestingly, the Winter-2030 scenario shows that the “Flexible cycle” performs worse than the steady-state “Annual cycle”. Higher net carbon removal costs for the “Flexible cycle” compared to the “Annual cycle” suggest that our method for optimizing the “Flexible cycle” is unsuitable for this scenario. The next chapter investigates in detail why the “Flexible cycle” operation performs worse than steady-state operation in the Winter-2030 scenario.

4.5 Limitations due to limited foresight

We examine the Winter-2030 | Medium-CapEx scenario in detail to analyze why the “Flexible cycle” performs worse than the “Annual cycle”. Figure 8a shows the electricity prices and GHG emissions. Unlike the Summer-2030 scenario, the 24-hour intervals driven by solar electricity are absent. Instead, unpredictable patterns are observed due to the high share of wind electricity. For example, some periods have nearly constant electricity

prices and GHG emissions, such as days two and three, while other periods show significant changes, like on the first day.

On the first day of Winter-2030, electricity prices and GHG emissions increase steadily (cf. Figure 8a). The “Flexible cycle” has higher net carbon removal costs than the “Annual cycle” during this period (cf. Figure 8b). The reason for the higher net carbon removal costs is the rolling horizon length of two cycles, with durations ranging from 4.5 to over 20 h. Hence, the net carbon removal costs after 4.5 h might be compared with those after 20 h for the process optimization. During upward trends in electricity prices and GHG emissions, this becomes problematic. When upward trends last longer than the minimum horizon length, short cycles are favored, as they are less affected by the rising input data. This is evident at the beginning of the Winter-2030 scenario. The first two cycles of the “Flexible cycle” have an average net carbon removal cost of 261 € t_{CO₂}⁻¹, compared to 291 € t_{CO₂}⁻¹ for the “Annual cycle”, showing an initial advantage (cf. Figure 8b). However, the “Flexible cycle” completes its first two cycles in 0.3 days, while the “Annual cycle” takes 0.6 days. After 0.3 days, the “Flexible cycle” still faces rising electricity prices and GHG emissions, whereas the “Annual cycle” has nearly passed these periods. Hence, during long periods of rising electricity prices and GHG emissions, a rolling horizon algorithm with a limited horizon length of two cycles fails to recognize that longer cycles perform better than short cycles because different points in time are compared.

The same pattern recurs from the middle of day 5 to the end of the time series (cf. Figure 8b). During this period, the “Flexible cycle” favors short cycles due to the upward trend in electricity prices and GHG emissions. By the mid of day 5, the “Flexible cycle” maintains an average net carbon removal cost that is about 4.8 % lower than the “Annual cycle”. However, due to the selection of short cycles during periods of rising electricity prices and GHG emissions, the “Flexible cycle” loses its average net carbon removal cost reduction by the end of day 7, resulting in an overall increase of 2.6 % for the entire week compared to the “Annual cycle”.

To test the hypothesis that limited foresight falsely favors short cycles, the “Flexible cycle” was re-optimized, now referred to as “Restricted Flexible cycle”. The minimum foresight was increased by setting the lower boundary for the adsorption and desorption phase times to 2.3 hours each (cf. Table 1). This extended the minimum foresight from 4.5 to about 9 hours. The adjustment allowed the “Restricted Flexible cycle” to reduce the average net carbon removal cost by almost 3.6 % compared to the “Annual cycle” and by 5.8 % compared to the normal

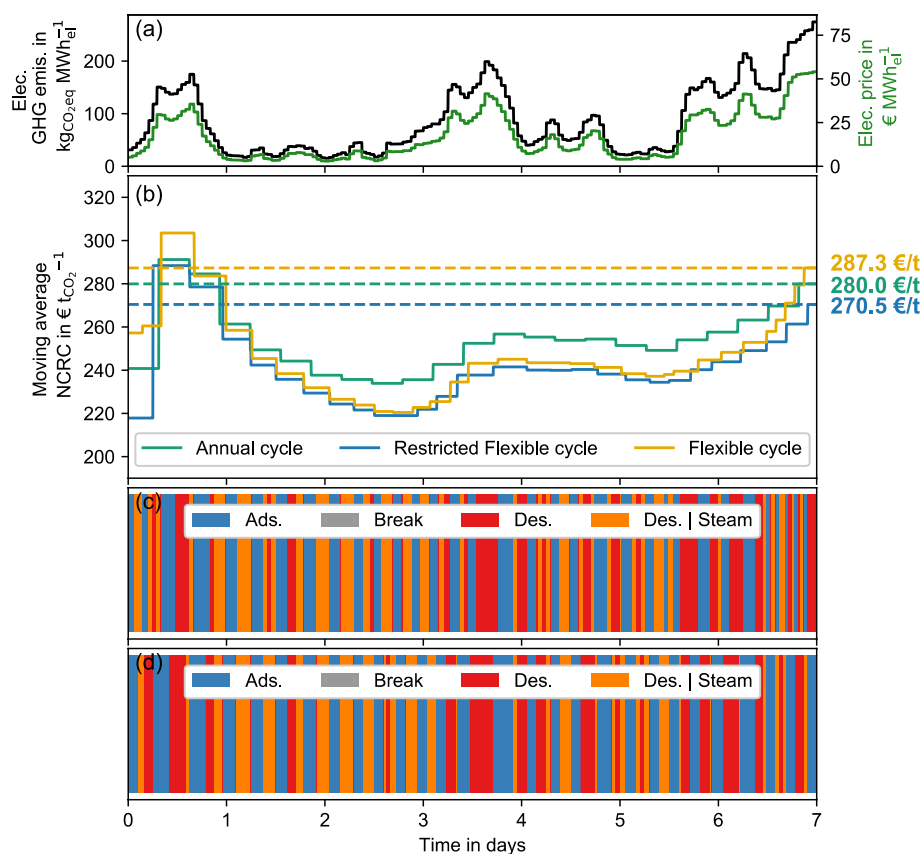


Fig. 8 Flexible DACCS operation for the Winter 2030 | Medium-CapEx scenario. **a** Inputs of the DACCS model: Electricity price in green and GHG emissions in black. **b** Moving average net carbon removal costs (NCRC) for three different operating modes: (i) the steady-state “Annual cycle”, (ii) “Flexible cycle”, and (iii) “Restricted Flexible cycle” with increased foresight. The dashed lines represent the respective weekly average net carbon removal costs for each operating mode. **c** Phases of the “Flexible cycle”, and **d** phases of the “Restricted Flexible cycle” providing insights into both flexible operation strategies. Note that the phases blowdown, heating, desorption, and cooling are summarized as desorption

“Flexible cycle” (cf. Figure 8b). Increasing foresight can thus prevent falsely selecting short cycles.

Even if increasing the lower boundaries of the adsorption and desorption phase times improves the average net carbon removal cost, it is not the best solution. The restriction limits the use of short cycles also when it would be beneficial. A more effective solution would be to increase the horizon length beyond two cycles to increase foresight. However, due to the complexity of our model and limited computational resources, increasing the horizon length beyond two cycles is not computationally feasible. Thus, our results are a lower estimate of the potential for flexible DACCS operation. The actual savings potential could be much higher. We hope our work inspires improvements in flexible DACCS methods to fully exploit cost reduction potential in the future.

5 Summary and conclusion

Flexible operation of direct air carbon capture and storage (DACCS) for demand side management is widely suggested to reduce the high costs of this energy-intensive technology. So far, limited studies provide process-specific options for a flexible operation, especially for adsorption-based DACCS using amine-functionalized sorbents. Our work aims to fill this gap. Starting from the standard steam-assisted temperature vacuum swing adsorption cycle, we propose measures to increase flexibility, including two break phases and variable air mass flow in the adsorption phase and steam flow in the desorption phase. To evaluate flexible operation, we use a DACCS system model combining detailed dynamic process modelling with life-cycle GHG emission and economic data.

The developed DACCS system model is optimized regarding the net carbon removal cost, representing the cost per net ton of CO₂ removed after accounting for all life-cycle GHG emissions. The modeled adsorption-based DACCS system uses electricity from a grid

with time-varying prices and GHG emissions attributable to its production. We analyze scenarios based on forecasted data for the German power grid in 2030 and 2040, considering different capital expenditures for a DACCS system with a nominal productivity of $4 \text{ kt yr}^{-1} \text{ CO}_2$. To optimize the process for flexible operation, we combine the DACCS system model with a rolling horizon algorithm. This algorithm allows optimizing the flexible operation for entire weeks while keeping the computational effort reasonable.

Our results show large reductions in net carbon removal cost through flexible operation. For a specific week in summer 2030, with capital expenditures of 7 € million, flexible operation reduces the average net carbon removal cost from $528.2 \text{ € t}_{\text{CO}_2}^{-1}$ to $488.1 \text{ € t}_{\text{CO}_2}^{-1}$, which is a 7.6 % reduction compared to steady-state operation. With a capital expenditure of 3 € million, the reduction is more pronounced, decreasing net carbon removal cost from $313 \text{ € t}_{\text{CO}_2}^{-1}$ to $275 \text{ € t}_{\text{CO}_2}^{-1}$ (-11.9 %). The largest reduction across all scenarios is observed for the summer week in 2040, characterized by high and highly fluctuating electricity prices. In this scenario, the flexible DACCS operation is particularly effective, achieving the highest reduction of 20.1 % in net carbon removal cost.

Overall, our work demonstrates the potential of flexible operation of adsorption-based DACCS to reduce costs and improve the economic viability. Future work should focus on refining optimization methods and exploring additional factors, e.g. weather or different sorbents, which influence flexible operation to fully exploit the cost-saving potential of adsorption-based DACCS.

Abbreviations

DSM	Demand-side management
GHG	Greenhouse gas
CDR	Carbon dioxide removal
DACCS	Direct air carbon capture and storage
DAC	Direct air carbon capture
TVSA	Temperature vacuum swing adsorption cycle
S-TVSA	Steam-assisted temperature vacuum swing adsorption cycle
CapEx	Capital expenditure
OpEx	Operational expenditure
HTHP	High-temperature heat pump
CRE	Carbon removal efficiency
GCRC	Gross carbon removal cost
NCRC	Net carbon removal cost
AF	Annuity factor

Supplementary Information

The online version contains supplementary material available at <https://doi.org/10.1007/s43979-025-00123-0>.

Supplementary Material 1.

Acknowledgements

Simulations were performed with computing resources granted by RWTH Aachen University under project *rwth0769*.

Authors' contributions

PP: conceptualization, data curation, formal analysis, investigation, software, methodology, visualization, writing – original draft, writing – review & editing. DR: conceptualization, software, formal analysis, investigation, methodology. ME: supervision, conceptualization, investigation, methodology, writing – review & editing. BN: conceptualization, data curation, writing – review & editing. NvdA: supervision, conceptualization, methodology, visualization, writing – review & editing. All authors read and approved the final manuscript.

Funding

This work has been carried out within the project MoGaTex (01LY2001B). The project is funded by the German Federal Ministry of Education and Research (BMBF) in the founding initiative “KMU-innovativ”.

Data availability

A detailed description of the dynamic DACCS system model can be found at <https://doi.org/https://doi.org/10.1039/D3EE02970K>. Additional details on the dynamic DACCS system model, performance evaluation, examined scenarios, and supplementary results are provided in the supporting information.

Declarations

Ethics approval and consent to participate

Not applicable.

Consent for publication

Not applicable.

Competing interests

There are no conflicts of interest to declare.

Received: 3 December 2024 Revised: 6 February 2025 Accepted: 19 February 2025

Published online: 26 March 2025

References

- IPCC (2022) Climate Change 2022: Mitigation of Climate Change. Contribution of Working Group III to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change. Cambridge University Press. <https://doi.org/10.1017/9781009157926.006>
- Buck HJ, Carton W, Lund JF, Markusson N (2023) Why residual emissions matter right now. *Nat Clim Change* 13(4):351–358. <https://doi.org/10.1038/s41558-022-01592-2>
- Steve Smith, Oliver Geden, Gregory F Nemet, Matthew Gidden, William F Lamb, Carter Powis, Rob Bellamy, Max Callaghan, Annette Cowie, Emily Cox, Sabine Fuss, Thomas Gasser, Giacomo Grassi, Jenna Greene, Sarah Lueck, Aniruddh Mohan, Finn Müller-Hansen, Glen Peters, Yoga Pratama, ... Jan Minx (2023) State of Carbon Dioxide Removal - 1st Edition. <https://doi.org/10.17605/OSF.IO/W3B4Z>
- Minx J, Probst B, Lueck S, Müller-Hansen F, Toetzel M, Diaz Anadon L, Engelmann T, Greene J, Hoffmann V, Nemet GF, Repke T (2024). Chapter 2: Research and Development. The State of Carbon Dioxide Removal - 2nd Edition. 2024. <https://doi.org/10.17605/OSF.IO/4FGHP>
- Fuss S, Canadell JG, Ciais P, Jackson RB, Jones CD, Lyngfelt A, Peters GP, van Vuuren DP (2020) Moving toward net-zero emissions requires new alliances for carbon dioxide removal. *One Earth* 3(2):145–149. <https://doi.org/10.1016/j.oneear.2020.08.002>
- Minx JC, Lamb WF, Callaghan MW, Fuss S, Hilaire J, Creutzig F, Amann T, Beringer T, de Oliveira Garcia W, Hartmann J, Khanna T, Lenzi D, Luderer G, Nemet GF, Rogelj J, Smith P, Vicente Vicente JL, Wilcox J, del Mar Zamora Dominguez, M. (2018) Negative emissions—Part 1: research landscape and synthesis. *Environ Res Lett* 13(6):63001. <https://doi.org/10.1088/1748-9326/aabf9b>

7. McQueen N, Gomes KV, McCormick C, Blumanthal K, Pisciotto M, Wilcox J (2021) A review of direct air capture (DAC): scaling up commercial technologies and innovating for the future. *Progress Energy* 3(3):32001. <https://doi.org/10.1088/2516-1083/abf1ce>
8. Bisotti F, Hoff KA, Mathisen A, Hovland J (2024) Direct Air capture (DAC) deployment: a review of the industrial deployment. *Chem Eng Sci* 283:119416. <https://doi.org/10.1016/j.ces.2023.119416>
9. Bouaboula H, Chaouki J, Belmabkhout Y, Zaabout A (2024) Comparative review of Direct air capture technologies: from technical, commercial, economic, and environmental aspects. *Chem Eng J* 484:149411. <https://doi.org/10.1016/j.cej.2024.149411>
10. Erans M, Sanz-Pérez ES, Hanak DP, Clulow Z, Reiner DM, Mutch GA (2022) Direct air capture: process technology, techno-economic and socio-political challenges. *Energy Environ Sci* 15(4):1360–1405. <https://doi.org/10.1039/D1EE03523A>
11. Fasihi M, Efimova O, Breyer C (2019) Techno-economic assessment of CO₂ direct air capture plants. *J. Clean. Prod.* 224(3):957–980. <https://doi.org/10.1016/j.jclepro.2019.03.086>
12. Sadiq A, Abdullatif Y, Aissa B, Ostovar A, Nassar N, El-Naas M, Amhamed A (2023) A review on progress made in direct air capture of CO₂. *Environ Technol Innov* 29(2):102991. <https://doi.org/10.1016/j.jeti.2022.102991>
13. Deutz S, Bardow A (2021) Life-cycle assessment of an industrial direct air capture process based on temperature–vacuum swing adsorption. *Nat Energy* 6(2):203–213. <https://doi.org/10.1038/s41560-020-00771-9>
14. Postweiler P, Engelpracht M, Rezo D, Gibelhaus A, von der Assen N (2024) Environmental process optimisation of an adsorption-based direct air carbon capture and storage system. *Energy Environ Sci* 18:2069. <https://doi.org/10.1039/D3EE02970K>
15. Terlouw T, Treyer K, Bauer C, Mazzotti M (2021) Life Cycle Assessment of Direct Air Carbon Capture and Storage with Low-Carbon Energy Sources. *Environ. Sci. Technol.* Advance online publication. <https://doi.org/10.1021/acs.est.1c03263>
16. Duffy P, Fitzpatrick C, Conway T, Lynch RP (2019) Issues in environmental science and technology: 46 // 5. Energy Storage Options and Their Environmental Impact // Energy storage options and their environmental impact // Energy Sources and Supply Grids – The Growing Need for Storage. The Royal Society of Chemistry. <https://doi.org/10.1039/9781788015530-00001>
17. Warren P (2014) A review of demand-side management policy in the UK. *Renew Sustain Energy Rev* 29(4):941–951. <https://doi.org/10.1016/j.rser.2013.09.009>
18. Gellings CW (1985) The concept of demand-side management for electric utilities. *Proceed IEEE* 73(10):1468–1470. <https://doi.org/10.1109/PROC.1985.13318>
19. Heffron R, Körner M-F, Wagner J, Weibelzahl M, Fridgen G (2020) Industrial demand-side flexibility: a key element of a just energy transition and industrial development. *Appl Energy* 269(4):115026. <https://doi.org/10.1016/j.apenergy.2020.115026>
20. Nilges B, Reinert C, von der Assen N (2024) Is demand-side management environmentally beneficial? Analyzing the greenhouse gas emissions due to load shifting in electric power systems. *J Clean Prod* 434:140062. <https://doi.org/10.1016/j.jclepro.2023.140062>
21. Breyer C, Fasihi M, Aghahosseini A (2020) Carbon dioxide direct air capture for effective climate change mitigation based on renewable electricity: a new type of energy system sector coupling. *Mitig Adapt Strateg Glob Chang* 25(1):43–65. <https://doi.org/10.1007/s11027-019-9847-y>
22. Singh U, Colosi LM (2022) Capture or curtail: The potential and performance of direct air capture powered through excess renewable electricity. *Energy Convers Manage* 215(6):100230. <https://doi.org/10.1016/j.ecmx.2022.100230>
23. van der Jagt S, Patankar N, Jenkins JD (2024) Understanding the role and design space of demand sinks in low-carbon power systems. *Energy Clim Change* 5(10):100132. <https://doi.org/10.1016/j.egycc.2024.100132>
24. Wohland J, Witthaut D, Schleussner C-F (2018) Negative emission potential of direct air capture powered by renewable excess electricity in Europe. *Earth's Futur* 6(10):1380–1384. <https://doi.org/10.1029/2018EF000954>
25. Alabi TM, Lawrence NP, Lu L, Yang Z, Bhushan Gopaluni R (2023) Automated deep reinforcement learning for real-time scheduling strategy of multi-energy system integrated with post-carbon and direct-air carbon captured system. *Appl Energy* 333(C):120633. <https://doi.org/10.1016/j.apenergy.2022.120633>
26. Arwa EO, Schell KR (2024) Batteries or silos: Optimizing storage capacity in direct air capture plants to maximize renewable energy use. *Appl Energy* 355:122345. <https://doi.org/10.1016/j.apenergy.2023.122345>
27. Wiegner JF, Grimm A, Weimann L, Gazzani M (2022) Optimal design and operation of solid sorbent direct air capture processes at varying ambient conditions. *Ind Eng Chem Res*, 176:acs.iecr.2c00681, 93. <https://doi.org/10.1021/acs.iecr.2c00681>
28. Liu Y, Miao Y, Feng Y, Wang L, Fujikawa S, Yu L (2025) Wind curtailment powered flexible direct air capture. *Appl Energy* 377:124402. <https://doi.org/10.1016/j.apenergy.2024.124402>
29. Merkert L, Harjunkoski I, Isaksson A, Säynevirta S, Saarela A, Sand G (2015) Scheduling and energy – Industrial challenges and opportunities. *Comput Chem Eng* 72(25):183–198. <https://doi.org/10.1016/j.compchemeng.2014.05.024>
30. Stampi-Bombelli V, van der Spek M, Mazzotti M (2020) Analysis of direct capture of CO₂ from ambient air via steam-assisted temperature–vacuum swing adsorption. *Adsorption* 26:1183–1197. <https://doi.org/10.1007/s10450-020-00249-w>
31. Young J, García-Díez E, García S, van der Spek M (2021) The impact of binary water–CO₂ isotherm models on the optimal performance of sorbent-based direct air capture processes. *Energy Environ Sci* 14(10):5377–5394. <https://doi.org/10.1039/D1EE01272J>
32. Postweiler P, Engelpracht M, Gibelhaus A, von der Assen N (2022) Environmental Process Optimisation of Adsorption-based Direct Air Capture of CO₂. 9th HPC Conference Proceedings, Vol. 9, 171–185. <https://heatpoweredcycles.eng.ed.ac.uk/sites/heatpoweredcycles.eng.ed.ac.uk/files/attachments/HPC2021-proceedings.pdf>
33. Bau U, Lanzerath F, Gräber M, Graf S, Schreiber H, Thielen N, Bardow A (2014) Adsorption energy systems library - Modeling adsorption based chillers, heat pumps, thermal storages and desiccant systems. In Linköping Electronic Conference Proceedings, Proceedings of the 10th International Modelica Conference, March 10–12, 2014. Lund, Sweden: Linköping University Electronic Press. pp. 875–883. <https://doi.org/10.3384/ecp14096875>
34. Institute of Technical Thermodynamics, RWTH Aachen University. (2022). SorpLib: Dynamic simulation of adsorption energy systems. <https://git.rwth-aachen.de/ltt/SorpLib>
35. Gebald C, Wurzbacher JA, Borgschulte A, Zimmermann T, Steinfeld A (2014) Single-component and binary CO₂ and H₂O adsorption of amine-functionalized cellulose. *Environ Sci Technol*. 48(4):2497–2504. <https://doi.org/10.1021/es404430g>
36. Wurzbacher JA, Gebald C, Brunner S, Steinfeld A (2016) Heat and mass transfer of temperature–vacuum swing desorption for CO₂ capture from air. *Chem Eng J*. 283(1):1329–1338. <https://doi.org/10.1016/j.cej.2015.08.035>
37. Wurzbacher JA, Gebald C, Piatkowski N, Steinfeld A (2012) Concurrent separation of CO₂ and H₂O from air by a temperature–vacuum swing adsorption/desorption cycle. *Environ Sci Technol*. 46(16):9191–9198. <https://doi.org/10.1021/es301953k>
38. Climeworks (2022) Orca is Climeworks' new large-scale carbon dioxide removal plant. <https://climeworks.com/roadmap/orca>
39. Young J, McQueen N, Charalambous C, Foteinis S, Hawrot O, Ojeda M, Pilorgé H, Andresen J, Psarras P, Renforth P, García S, van der Spek M (2022). The cost of direct air capture and storage: the impact of technological learning, regional diversity, and policy. <https://doi.org/10.26434/chemrxiv-2022-dp36t-v2>
40. DIN V 18599-7, Deutsches Institut für Normung e. V (2011) Energy efficiency of buildings -Calculation of the net, final and primary energy demand for heating, cooling, ventilation, domestic hot water and lighting: Part 7: Final energy demand of air-handling and air-conditioning systems for non-residential buildings (DIN V 18599-7)
41. Baumgärtner N, Deutz S, Reinert C, Nolzen N, Kuepper LE, Hennen M, Hollermann DE, Bardow A (2021) Life-cycle assessment of sector-coupled national energy systems: environmental impacts of electricity, heat, and transportation in Germany Till 2050. *Front Energy Res* 9:621502, 886. <https://doi.org/10.3389/fenrg.2021.621502>
42. Grossmann IE (2012) Advances in mathematical programming models for enterprise-wide optimization. *Comput Chem Eng* 47:2–18. <https://doi.org/10.1016/j.compchemeng.2012.06.038>

43. Glomb L, Liers F, Rösel F (2022) A rolling-horizon approach for multi-period optimization. *Eur J Oper Res* 300(1):189–206. <https://doi.org/10.1016/j.ejor.2021.07.043>
44. Functional Mock-up Interface (2022). <https://fmi-standard.org/>
45. Blank J, Deb K (2020) Pymoo: multi-objective optimization in python. *IEEE Access* 8:89497–89509. <https://doi.org/10.1109/ACCESS.2020.2990567>
46. Zhan ZH, Zhang J, Li Y, Chung HSH (2009) Adaptive particle swarm optimization. *IEEE Transactions on Systems, Man, and Cybernetics. Part B, Cybernetics: A Publication of the IEEE Systems, Man, and Cybernetics Society*, 39(6):1362–1381. <https://doi.org/10.1109/TSMCB.2009.2015956>
47. Rezo D, Postweiler P, Engelpracht M, Meuleneers L, von der Aßen N (2024) A method for siting adsorption-based direct air carbon capture and storage plants for maximum CO2 removal. *Carbon Neutral*, 3(1). <https://doi.org/10.1007/s43979-024-00100-z>
48. Sendi M, Bui M, Mac Dowell N, Fennell P (2022) Geospatial analysis of regional climate impacts to accelerate cost-efficient direct air capture deployment. *One Earth* 5(10):1153–1164. <https://doi.org/10.1016/j.oneear.2022.09.003>
49. Klima Aachen (2022). <https://de.climate-data.org/europa/deutschland/nordrhein-westfalen/aachen-2115/#climate-table>
50. Nilges B, Haese A, Reinert C, von der Assen N (2025) Validity ranges of locational marginal emission factors for the environmental assessment of electric load shifting. *Appl Energy* 380:125069. <https://doi.org/10.1016/j.apenergy.2024.125069>
51. Reinert C, Schellhas L, Mannhardt J, Shu DY, Kämper A, Baumgärtner N, Deutz S, Bardow A (2022) SecMOD: an open-source modular framework combining multi-sector system optimization and life-cycle assessment. *Front Energy Res* 10(884525):1038. <https://doi.org/10.3389/fenrg.2022.884525>
52. UNESCO (2023) Carbon capture and storage plant becomes operational in Iceland. <https://www.unesco.org/en/articles/carbon-capture-and-storage-plant-becomes-operational-iceland>
53. Schellevis M (2023) CO2 capture from air: a process engineering approach. University of Twente
54. Wang J, Li S, Deng S, Zeng X, Li K, Liu J, Yan J, Lei L (2023) Energetic and life cycle assessment of direct air capture: a review. *Sustain Prod Consum* 36:1–16. <https://doi.org/10.1016/j.spc.2022.12.017>

Publisher's Note

Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.